A FLAT FINITE TYPE ALGEBRA OVER A NOETHERIAN RING WITH GEOMETRICALLY INTEGRAL FIBRES IS PROJECTIVE AS A MODULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Lifting projectivity along a nilpotent ideal. Algebra, label algebra-lemma-lift-projective, invoked at proof line 2202. Shows that if S/JS is projective over R/J with J the radical of I, then S/IS is projective over R/I because J/I is nilpotent and S/IS is flat, contradicting the maximality of I; hence I may be taken radical.
- Generic flatness for a finite type algebra over a reduced Noetherian ring. Algebra, labels algebra-lemma-generic-flatness-Noetherian (proof line 2212) and algebra-lemma-generic-flatness-reduced (proof line 2214). Produces a nonzero f with S_f free over R_f, which enables the faithfully flat presentation of R by R_f times its f-adic completion.
- Faithfully flat descent of projectivity, flatness of a completion, and the criterion for a flat pure module over a complete local ring. Algebra, labels algebra-lemma-completion-flat (proof line 2222) and algebra-theorem-ffdescent-projectivity (proof line 2224); Same chapter flat.tex, label lemma-flat-pure-over-complete-projective (proof line 2227). Reduces projectivity of S over R to projectivity of the completed base change, and the criterion then applies because the fibre rings stay geometrically integral and the element f lies in the Jacobson radical of the complete ring.

Source reference. Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Navigation pointer for \ref{algebra-lemma-completion-flat} used inside the counted span (line 2222); the argument that consumes it is retained in the span.

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Source reference. Original lemma 051R: sources/stacks-chow/tags.txt, source lines 6544–6544. Navigation pointer for \ref{algebra-lemma-generic-flatness-Noetherian} used inside the counted span (line 2212); the argument that consumes it is retained in the span.

Source reference. Original lemma 051Z: sources/stacks-chow/tags.txt, source lines 6552–6552. Navigation pointer for \ref{algebra-lemma-generic-flatness-reduced} used inside the counted span (line 2214); the argument that consumes it is retained in the span.

Source reference. Original lemma 05CG: sources/stacks-chow/tags.txt, source lines 6919–6919. Navigation pointer for \ref{algebra-lemma-lift-projective} used inside the counted span (line 2202); the argument that consumes it is retained in the span.

Source reference. Original lemma 05A9: sources/stacks-chow/tags.txt, source lines 6842–6842. Navigation pointer for \ref{algebra-theorem-ffdescent-projectivity} used inside the counted span (line 2224); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DP: sources/stacks-flat/pinned-original-source.tex, source lines 2145–2145. Navigation pointer for \ref{lemma-flat-pure-over-complete-projective} used inside the counted span (line 2227); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $R \to S$ be a ring map. Assume*

(1) *R is Noetherian,*
(2) *$R \to S$ is of finite type and flat, and*
(3) *every fibre ring $S \otimes_R \kappa(\mathfrak{p})$ is geometrically integral over $\kappa(\mathfrak{p})$.*

Then S is projective as an R-module.

Proof. Consider the set

$$\{I \subset R \mid S/IS \text{ not projective as } R/I\text{-module}\}$$

We have to show this set is empty. To get a contradiction assume it is nonempty. Then it contains a maximal element I. Let $J = \sqrt{I}$ be its radical. If $I \neq J$, then S/JS is projective as a R/J-module, and S/IS is flat over R/I and J/I is a nilpotent ideal in R/I. Applying Algebra, Lemma 05CG we see that S/IS is a projective R/I-module, which is a contradiction. Hence we may assume that I is a radical ideal. In other words we are reduced to proving the lemma in case R is a reduced ring and S/IS is a projective R/I-module for every nonzero ideal I of R.

Assume R is a reduced ring and S/IS is a projective R/I-module for every nonzero ideal I of R. By generic flatness, Algebra, Lemma 051R (applied to a localization R_g which is a domain) or the more general Algebra, Lemma 051Z there exists a nonzero $f \in R$ such that S_f is free as an R_f-module. Denote $R^\wedge = \lim R/(f^n)$ the (f)-adic completion of R. Note that the ring map

$$R \longrightarrow R_f \times R^\wedge$$

is a faithfully flat ring map, see Algebra, Lemma 00MB. Hence by faithfully flat descent of projectivity, see Algebra, Theorem 05A9 it suffices to prove that $S \otimes_R R^\wedge$ is a projective $R^\wedge$-module. To see this we will use the criterion of Lemma 05DP. First of all, note that $S/fS = (S \otimes_R R^\wedge)/f(S \otimes_R R^\wedge)$ is a projective $R/(f)$-module and that $S \otimes_R R^\wedge$ is flat and of finite type over $R^\wedge$ as a base change of such. Next, suppose that $\mathfrak{p}^\wedge$ is a prime ideal of $R^\wedge$. Let $\mathfrak{p} \subset R$ be the corresponding prime of R. As $R \to S$ has geometrically integral fibre rings, the same is true for the fibre rings of any base change. Hence $\mathfrak{q}^\wedge = \mathfrak{p}^\wedge(S \otimes_R R^\wedge)$, is a prime ideals lying over $\mathfrak{p}^\wedge$ and it is the unique associated prime of $S \otimes_R \kappa(\mathfrak{p}^\wedge)$. Thus

we win if $f(S \otimes_R R^\wedge) + \mathfrak{q}^\wedge \neq S \otimes_R R^\wedge$. This is true because $\mathfrak{p}^\wedge + fR^\wedge \neq R^\wedge$ as f lies in the Jacobson radical of the f-adically complete ring $R^\wedge$ and because $R^\wedge \to S \otimes_R R^\wedge$ is surjective on spectra as its fibres are nonempty (irreducible spaces are nonempty). $\square$